Danmarks
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02435 Decision Making Under Uncertainty
Assignment Part A - Group 20

AUTHORS

Niklas William Ward Petersen
StudentID: s215215
Theodor Dornonville de la Cour
StudentID: s225093
Fabian Mengedoht
StudentID: s226530

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GENAI DECLARATION

We have used artificial intelligence tools from OpenAI, Anthropic, and Google for coding and to assist with the clarity and grammatical structure of the report. In accordance with DTU's guidelines for documenting the use of Generative AI, we acknowledge this usage and can provide documentation including prompts, AI responses, and version information upon request.

TASK 1: OPTIMAL IN HINDSIGHT SOLUTION

Sets

- $\mathcal{T} : \{0, \ldots, 9\}$: Set of hours in the daily operation.
- $\mathcal{R} : \{0, 1\}$: Set of rooms in the restaurant.

Indices

- $t \in \mathcal{T}$: Time index for the hour of the operation-day
- $r \in \mathcal{R}$: Room index for the room of the restaurant

Decision variables

- v_t: Status of ventilation system in time t, $v_t \ \epsilon \ \{0, 1\}$
- $p_{r,t}$: Heating power of heater in time t and room r, $0 \leq p_{r,t} \leq \overline{\mathrm{P}}_r$
- $t_{r,t}$: Temperature in room r at hour t
- h_t: Humidity at hour t, $0 \leq h_t \leq 100$
- s_t: start-up variable to model inertia, at hour t, $s_t \ \epsilon \ \{0, 1\}$
- $z_{r,t}^{high}$: If heater is forced to operate at zero power in room r at hour t due to the high temperature overrule, $z_{r,t}^{high} \ \epsilon \ \{0, 1\}$
- $z_{r,t}^{low}$: If heater is forced to operate at full power in room r at hour t due to the low temperature overrule, $z_{r,t}^{low} \ \epsilon \ \{0, 1\}$

Parameters

- $\overline{\mathrm{P}}_r$: Upper limit of heating power provided by the heating unit.
- $\widetilde{\lambda}_t$: Electricity price for each hour t.
- P^{vent}: Power Usage of central ventilation when tuned on.
- $\mathrm{O}_{r,t}$: Occupancy in the room's r at hour t (number of people).
- T_t^{out}: Outside temperature at hour t.
- T^{low}:The room's temperature lower-bound threshold.
- T^{ok}: The temperature where the rooms temperature is okay.
- T^{high}: The room's temperature upper-bound threshold.
- H^{high}: The room's high Humidity.
- ζ^{exch}: Thermal heat exchange between the two rooms at $t-1$.
- ζ^{loss}: The heat losses through walls of the rooms.
- ζ^{conv}: Power of the room's heater at$t-1$.
- ζ^{cool}: The ventilation status at $t-1$.
- ζ^{occ}: Coefficient representing the impact of the occupancy level at time $t-1$ on room's temperature.
- η^{occ}: Coefficient representing the impact of the occupancy level at time $t-1$ on the room's humidity.
- η^{vent}: Ventilation status at hour $t-1$.

Objective Function

The objective is to reduce the restaurants total cost of electricity during the day. Mathematically this is expressed as:

$$\text{Minimize } Z = \sum_{t \in \mathcal{T}} (\widetilde{\lambda}_t * (\mathrm{P}^{vent} * v_t + \sum_{r \in \mathcal{R}} p_{r,t}))$$

We sum up all the kw(h) used during each period y the vent and the heaters in both rooms and multiply this by the price per kw(h)

Constraints

<u>Humidity Dynamics:</u>

We set an initial humidity, as the humidity transition constraint uses info from the previous time period and we are given an initial humidity

$$h_0 = initial\ humidity \tag{1}$$

This is the linear function of how the humidity changes to the next period as specified in the assignment

$$h_t = h_{t-1} + \eta^{occ} * \sum_{r \in \mathcal{R}} \mathrm{O}_{r,t-1} - \eta^{vent} * v_{t-1} \quad \forall t \ \epsilon \ \mathcal{T} \setminus \{0\} \tag{2}$$

<u>Temperature Dynamics:</u>

We set an initial temperature, as the temperature transition constraint uses info from the previous time period and we are given an initial temperature

$$t_{r,0} = initial\ temperature\ of\ room\ r \quad \forall r \in \mathcal{R} \tag{3}$$

This is the linear function of how the temperature changes to the next period as specified in the assignment with one constraint per room.

$$t_{0,t} = t_{0,t-1} + \zeta^{exch} * (t_{1,t-1} - t_{0,t-1}) + \zeta^{loss} * (\mathrm{T}^{out}_{t-1} - t_{0,t-1}) + \zeta^{conv} * p_{0,t-1} - \zeta^{cool} * v_{t-1} + \zeta^{occ} * \mathrm{O}_{0,t-1} \quad \forall t \in \mathcal{T} \setminus \{0\} \tag{4}$$

$$t_{1,t} = t_{1,t-1} + \zeta^{exch} * (t_{0,t-1} - t_{1,t-1}) + \zeta^{loss} * (\mathrm{T}^{out}_{t-1} - t_{1,t-1}) + \zeta^{conv} * p_{1,t-1} - \zeta^{cool} * v_{t-1} + \zeta^{occ} * \mathrm{O}_{1,t-1} \quad \forall t \in \mathcal{T} \setminus \{0\} \tag{5}$$

Low-temperature overrule:

This constraint makes sure that low-temp overrule controller ($z^{low}_{r,t}$) is activated when the temperature drops below T^{low}

$$\mathrm{T}^{low} - t_{r,t} \leq \mathrm{M} * z^{low}_{r,t} \quad \forall r \in \mathcal{R},\ t \in \mathcal{T} \tag{6}$$

This constraint makes sure that if the overrule controller was active in the period before and the temperature is not above T^{OK} yet, it will keep the overrule controller active

$$z^{low}_{r,t} \geq z^{low}_{r,t-1} - 1 + \frac{\mathrm{T}^{OK} - t_{r,t}}{\mathrm{M}} \quad \forall r \in \mathcal{R},\ t \in \mathcal{T} \setminus \{0\} \tag{7}$$

This is constraint makes sure that if the overrule controller is active, that the heater will actually operate on full power. It also naturally acts as non-negativity constraint for $p_{r,t}$

$$p_{r,t} \geq z^{low}_{r,t} * \overline{\mathrm{P}}_r \quad \forall r \in \mathcal{R},\ t \in \mathcal{T} \tag{8}$$

High-temperature overrule:

Works the same way as the constraint in equation 6, but activating the high-temp overrule if temperature goes above T^{high}

$$t_{r,t} - \mathrm{T}^{high} \leq \mathrm{M} * z^{high}_{r,t} \quad \forall r \in \mathcal{R},\ t \in \mathcal{T} \tag{9}$$

This makes sure that if the overrule is active the heaters operate at zero power. This also naturally acts as capacity constraint

$$p_{r,t} \leq (1 - z^{high}_{r,t}) * \overline{\mathrm{P}}_r \quad \forall r \in \mathcal{R},\ t \in \mathcal{T} \tag{10}$$

Humidity Overrule Controller:

Similar to the rules before, This will make sure that the ventilation is on, if the humidity goes above a certain percentage

$$h_t \leq \mathrm{H}^{high} + \mathrm{M} * v_t \quad \forall t \in \mathcal{T} \tag{11}$$

Ventilation System Inertia:

We set the initial start-up variable to the initial ventilation period, as there are no other periods before that could influence this.

$$v_0 = s_0 \tag{12}$$

This forces the start-up variable to be 1 if the ventilation was turned on in t. For periods after that, it will default to zero

$$s_t \geq v_t - v_{t-1} \quad \forall t \in \mathcal{T} \setminus \{0\} \tag{13}$$

This makes sure that if the inertia set in, the ventilation will be on for at least 3 periods

$$v_t + v_{t+1} + v_{t+2} \geq 3 * s_t \quad \forall t \in \mathcal{T} \setminus \{8, 9\} \tag{14}$$

Suitable number for Big M:

The Big M is used in multiple constraints related to temperature differences. It should be as large as the maximum temperature difference we would observe. Conservatively speaking, for Copenhagen for example we can say the lowest temp could be -30 and the highest 40, giving us a Big M of 70. For the Big M in the humidity constraint it is sufficient to take the difference between the maximum (100%) and H^{high} from the overrule rule.

Python

We handed in the Python script (Jupyter notebook) as an appendix to this pdf. Please note that we slightly adjusted the code of the given Plots file, but the formatting stayed the same. We merely adjusted it so we could use it in our model. The adjusted code of the plots file is in our appendix as well. Additionally, please note that we import the data from the given "SystemCharacteristics.py" file.

Results

We found an average daily electricity cost over the 100 days of 111.39€.
In figure 1 at the end we can see the results of day 0 from the dataset. In order for the humidity to to rise above the high threshold it has to run at least a few times a day as we can see from the humidity graph that reaches almost 70 at the end of the day. The is the only way to decrease the humidity and it uses the cheap electricity in the middle of the day to start the ventilation. It runs for 4 periods in a row, showcasing the inertia of the system. We can se th effect of the Occupancy on heat nicely in period 1. No heating was done in period 0 and since room 1 was fuller in period 0, it has a higher temperature in period 1 leading to a slightly higher heating power of room 2 in period 1. We also see from the temperature curve that the temp generally tries to be in between the low and high overrule temperature. Also the heaters also take advantage of the cheap electricity during the day.
For day 1 in figure 2 we see the same patterns. Here the electricity price is cheaper earlier in the day, therefore most of the heating and ventilation is at that time. To conclude the main takeaways are that it makes sense to try to stay within the limits of the overrule controllers and and try to heat and dehumidify the room during periods with cheap electricity. Generally we see that the difference between room temperatures and the occupancy of the rooms have a noticeable effect. In the sample days the outdoor temperature has the biggest effect, has they are around zero and there is a high temperature delta. (See rapid decline in temperature at end of day 1)

TASK 2: MARKOV DECISION PROCESS

Sets

- $\mathcal{R} : \{1, 2\}$: Set of rooms.
- $\mathcal{T} : \{1, \ldots, 10\}$: Set of hours of the restaurant's operation.

Note for the Set of hours of operation we are switching from index zero to one as we will not model the MDP in python.

Stages

- $t \in \mathcal{T}$: Hour (stage) index.

State Variables

The system's state at hour t is denoted by x_t and includes all variables required to describe the current physical conditions and contextual information:

$$x_t = \left(T_{1,t}, T_{2,t}, H_t, O_{1,t}, O_{2,t}, v_{t-1}, v_{t-2}, v_{t-3}, z^{low}_{1,t-1}, z^{low}_{2,t-1}, \widetilde{\lambda}_t, \widetilde{\lambda}_{t-1}\right)$$

where:

- $T_{r,t}$: Temperature of room r at hour t.
- H_t: Humidity of the restaurant at hour t.
- $O_{r,t}$: Occupancy level of room r at hour t.
- v_{t-1}: Ventilation status at previous hour. (1 if ON, 0 if OFF).
- v_{t-2}: Ventilation status two hours ago. (1 if ON, 0 if OFF).
- v_{t-3}: Ventilation status three hours ago. (1 if ON, 0 if OFF).
- $z^{low}_{r,t-1}$: Temperature overrule controller for low temperature in previous period. (1 if ON, 0 if OFF).
- $\widetilde{\lambda}_t$: Electricity price at hour t
- $\widetilde{\lambda}_{t-1}$: Electricity Price from previous hour

Decision Variables

The decision (or action) at hour t is denoted by u_t:

$$u_t = (p_{1,t}, p_{2,t}, v_t)$$

where:

- $p_{r,t}$: Heating power assigned to the heater of room r at hour t, with bounds $0 \leq p_{r,t} \leq \overline{P}_r$.
- $v_t \in \{0, 1\}$: Ventilation status (1 if ON, 0 if OFF).

Transition Dynamics

Fig. 1: Results for day 0

The temperature and humidity are governed by fixed system parameters, making them fully deterministic, their dynamics are therefore identical to the equality constraints in the MILP.

Temperature Dynamics: For each room $r \in \mathcal{R}$, letting r' denote the other room.

$$T_{r,t+1} = T_{r,t} + \zeta^{\text{exch}}(T_{r',t} - T_{r,t}) + \zeta^{\text{loss}}\left(\mathrm{T}_t^{\text{out}} - T_{r,t}\right) + \zeta^{\text{conv}}\, p_{r,t} - \zeta^{\text{cool}}\, v_t + \zeta^{\text{occ}}\, O_{r,t} \tag{15}$$

Humidity Dynamics:

$$H_{t+1} = H_t + \eta^{\text{occ}} \sum_{r \in \mathcal{R}} O_{r,t} - \eta^{\text{vent}}\, v_t \tag{16}$$

The electricity price and occupancy follow a given stochastic processes as specified in Part B of the assignment.

Occupancy Dynamics:

The occupancy of each room at $t+1$ is a function of the current occupancy of both rooms, capturing the correlation between rooms:

$$O_{r,t+1} \sim P(\cdot \mid O_{r,t},\ O_{r',t}) \tag{17}$$

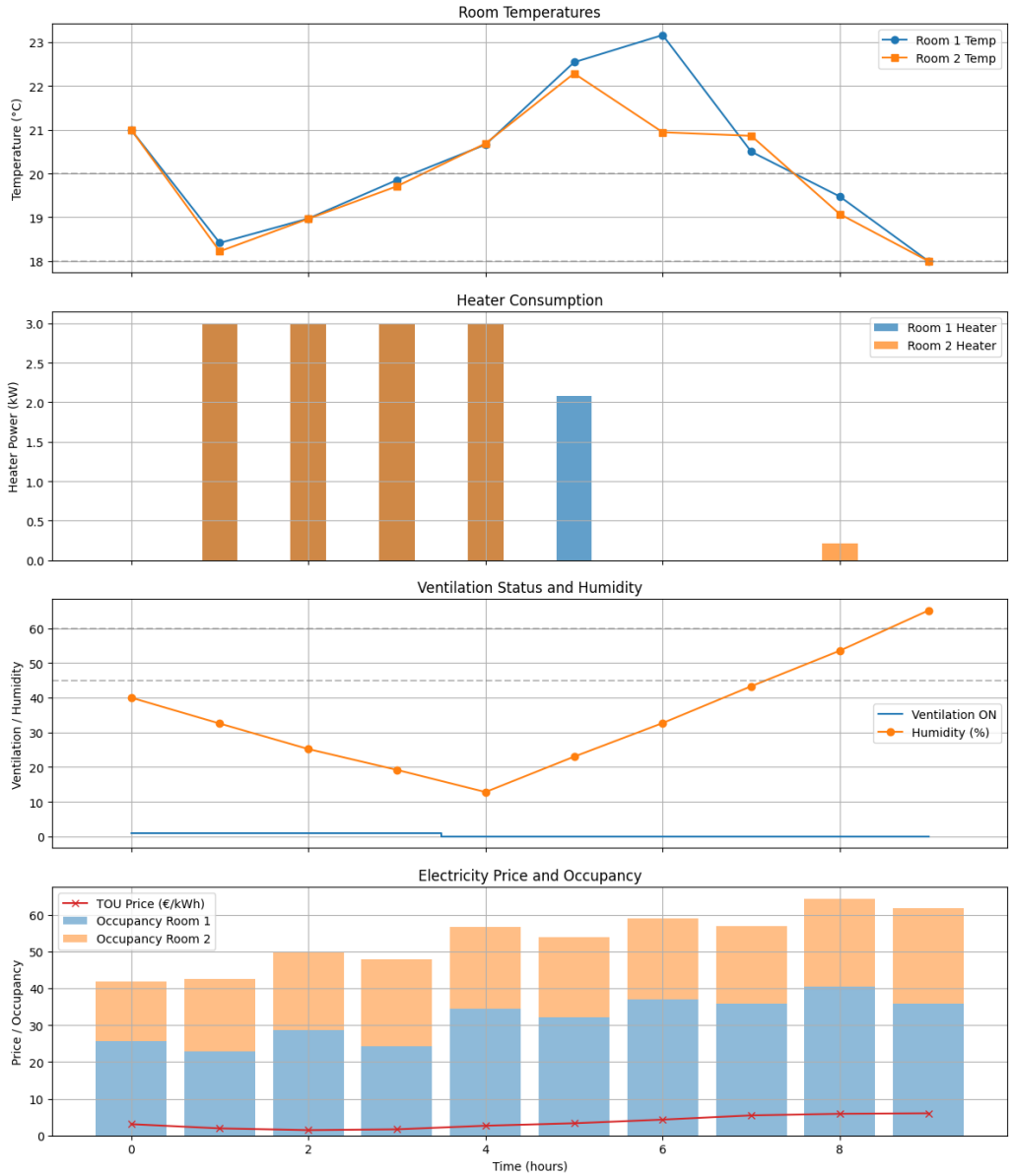

Fig. 2: Results for day 1

Electricity Price:

The electricity price at $t+1$ depends on the two prices before:

$$\widetilde{\lambda}_{t+1} \sim P\Big(\cdot \mid \widetilde{\lambda}_t,\ \widetilde{\lambda}_{t-1}\Big) \tag{18}$$

Auxiliary constraints:

We use some of the known parameters from our original MILP model. These constraints are modeling the overrule and capacity constraints as seen in the MILP, as well as the inertia of the ventilation system. We cannot influence these dynamics, but our decision are influenced/bound by them.

$$z_{r,t}^{low} = \begin{cases} 1, & \text{if } T_{r,t} \leq T^{low} \\ z_{r,t-1}^{low} & \text{if } T^{low} < T_{r,t} < T^{OK} \\ 0, & \text{if } T_{r,t} \geq T^{OK} \end{cases} \tag{19}$$

$$p_{r,t} = \begin{cases} \overline{\mathrm{P}}_r & \text{if } z_{r,t}^{low} = 1 \\ 0 & \text{if } T_{r,t} \geq T^{high} \\ [0, \overline{P}_r] & otherwise \end{cases} \tag{20}$$

$$v_t = \begin{cases} 1 & \text{if } H_t \geq \mathrm{H}^{high} \\ 1 & \text{if } v_{t-1} = 1 \text{ and } v_{t-2} = 0 \\ 1 & \text{if } v_{t-1} = 1 \text{ and } v_{t-2} = 1 \text{ and } v_{t-3} = 0 \\ \{0, 1\} & otherwise \end{cases} \tag{21}$$

Cost Function

The stage cost represents the electricity expenditure at hour t, equal to the realized electricity price multiplied by the total power consumption from heating and ventilation.

$$c(x_t, u_t) = \widetilde{\lambda}_t \left(\sum_{r \in \mathcal{R}} p_{r,t} + v_t P^{\text{vent}} \right) \tag{22}$$

GROUP PARTICIPATION

A. Group statement

We believe that we have contributed equally and would distribute the contribution as follows: $f_i = \frac{1}{3} \ \forall i \, \epsilon \, \{Niklas, Theodor, Fabian\}$ We have split the task at first and worked on it ourselves during the first few exercise sessions, and later helped each other with the remaining work. We all verified and checked each others work

B. Niklas statement

I was responsible for developing the nomenclature for both Task 1 and Task 2. My primary focus was on formulating the MDP model and assisting in specifying the transition dynamics of the MILP formulation. Additionally, I contributed some debugging in Python and general report writing/editing.

C. Theodors statement

I worked a bit on the MILP model in the beginning and then started with my primary focus, the python implementation. Additionally I helped with the MDP formulation and did some report correction.

D. Fabians statement

I primarily worked on the Task 1. I designed most of the MILP model and helped turning it into the Python script and validified and testing the answers using the given plots. For Task 2 I helped with some variable declaration and added auxiliary constraints.